```
1000 'THIS SUBROUTINE IS CALLED FROM FIG. 33-7,  LINE 340
1010 '
1020 'Variables entering subroutine:      PI, FC, LH, PR, HP, P%
1030 'Variables exiting subroutine:       A0, A1, A2, B1, B2
1040 'Variables used internally:          RP, IP, ES, VX, KX, T, W, M, D, K,
1050 '                                    X0, X1, X2, Y1, Y2
1060 '
1070 '                                    'Calculate pole location on unit circle
1080 RP = -COS(PI/(NP*2) + (P%-1) * PI/NP)
1090 IP =  SIN(PI/(NP*2) + (P%-1) * PI/NP)
1100 '
1110 '                                    'Warp from a circle to an ellipse
1120 IF PR = 0 THEN GOTO 1210
1130 ES = SQR( (100 / (100-PR))^2 -1 )
1140 VX = (1/NP) * LOG( (1/ES) + SQR( (1/ES^2) + 1) )
1150 KX = (1/NP) * LOG( (1/ES) + SQR( (1/ES^2) - 1) )
1160 KX = (EXP(KX) + EXP(-KX))/2
1170 RP = RP * ( (EXP(VX) - EXP(-VX) ) /2 ) / KX
1180 IP  = IP * ( (EXP(VX) + EXP(-VX) ) /2 ) / KX
1190 '
1200 '                                    's-domain to z-domain conversion
1210 T  = 2 * TAN(1/2)
1220 W  = 2*PI*FC
1230 M  = RP^2 + IP^2
1240 D = 4 - 4*RP*T + M*T^2
1250 X0 = T^2/D
1260 X1 = 2*T^2/D
1270 X2 = T^2/D
1280 Y1 = (8 - 2*M*T^2)/D
1290 Y2 = (-4 - 4*RP*T - M*T^2)/D
1300 '
1310 '                                    'LP TO LP, or LP TO HP
1320 IF LH = 1 THEN K = -COS(W/2 + 1/2) / COS(W/2 - 1/2)
1330 IF LH = 0 THEN K =  SIN(1/2 - W/2) / SIN(1/2 + W/2)
1340 D = 1 + Y1*K - Y2*K^2
1350 A0 = (X0 - X1*K + X2*K^2)/D
1360 A1 = (-2*X0*K + X1 + X1*K^2 - 2*X2*K)/D
1370 A2 = (X0*K^2 - X1*K + X2)/D
1380 B1 = (2*K + Y1 + Y1*K^2 - 2*Y2*K)/D
1390 B2 = (-K^2 - Y1*K + Y2)/D
1400 IF LH = 1 THEN A1 = -A1
1410 IF LH = 1 THEN B1 = -B1
1420 '
1430 RETURN
```

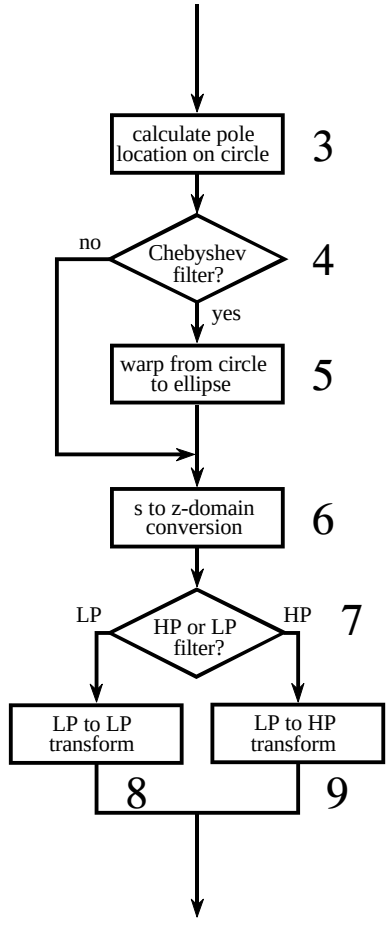

FIGURE 33-8
Subroutine called from Figure 33-7.

from the s-plane to the z-plane, such as vertical lines being mapped into circles. Here is an example of how it works. For a continuous system with a single pole-pair located at $p_1 = \sigma + j\omega$ and $p_2 = \sigma - j\omega$, the s-domain transfer function is given by:

$$H(s) = \frac{1}{(s - p_1)(s - p_2)}$$

The bilinear transform converts this into a discrete system by replacing each *s* with the expression given in Eq. 33-10. This creates a z-domain transfer